

REPRESENTATION CALCULUS FOR THETA-EXPECTATIONS

QIAN QI

ABSTRACT. This paper develops the representation theory that lies downstream of the first-principles theta-expectation construction. The nonlinear semigroup and its nonconvex HJB generator are taken as outputs of the companion homogenization paper. After this input is fixed, each sufficiently regular payoff determines a linearized parabolic generator, a *carre-du-champ*, a calibrated diffusion representation, a decoupled BSDE, and a semigroup-level Cameron-Martin–Girsanov formula. These objects represent the already-derived semigroup; they are not inputs to the deterministic billiard derivation.

1. INTRODUCTION

This is the third article in the sequence. Paper 1 proves the singular billiard response package. Paper 2 uses that package to derive the nonconvex HJB equation and the theta-expectation semigroup from deterministic billiards. The present paper begins only after those two steps are complete.

Representation, not derivation. The central point is the payoff dependence of every classical probabilistic representation. A genuinely nonlinear HJB semigroup cannot be represented by a single classical martingale problem for all test functions. Once a terminal payoff is fixed, however, its decoupling field freezes the nonlinear generator into an ordinary linear parabolic operator. Diffusions, BSDEs, and Girsanov formulae are then available for that calibrated linear problem.

1.1. Theorem map.

Layer	Paper theorem	Proof role
Semigroup input	Assumption 2.1	imported from Paper 2
Obstruction	Theorem 3.1	rules out one universal law
Linearization	Theorem 4.2	freezes a fixed payoff
Regularized diffusion	Theorem 5.2	represents the calibrated linear problem
BSDE	Theorem 6.2	uses $Z = \sigma^\top \nabla u$
Orientation	Proposition 7.1	checks terminal-value signs
Girsanov	Theorem 8.2	semigroup-level shift identity
Stress test	Proposition 9.2	short application illustration

1.2. Provenance and dependency guardrail. The stochastic-analysis tools used below are classical: the martingale-problem representation follows [6], the parabolic and viscosity stability language follows [1, 2], the BSDE convention follows [3], and the stochastic-calculus/Girsanov notation follows [5]. Nonlinear expectation terminology follows [4]. These references are used only after the deterministic theta-expectation semigroup has been derived in the companion HJB paper.

2. INPUT FROM THE THETA-EXPECTATION PAPER

Assumption 2.1 (Derived nonlinear semigroup). The companion theta-expectation paper has constructed coefficients

$$D(x, p) \in \mathbb{R}_{\text{sym},+}^{d \times d}, \quad H(x, p) \in \mathbb{R},$$

from deterministic billiard response theory and finite-response mechanics. It has also proved comparison and uniqueness for the terminal-value HJB equation

$$(1) \quad -\partial_t u - \operatorname{tr}(D(x, \nabla u) \nabla^2 u) - H(x, \nabla u) = 0, \quad u(T, \cdot) = \phi.$$

The normalized semigroup is

$$(2) \quad \mathcal{E}_{t,T}^\theta[\phi](x) = u(t, x).$$

It is constant preserving, cash additive, monotone, and time consistent, and it is generally nonlinear and not subadditive.

Remark 2.2. Assumption 2.1 is the only connection to deterministic billiards used in this paper. All microscopic response estimates and the HJB homogenization theorem belong to the companion papers.

Assumption 2.3 (Representation window). Fix a compact cylinder $Q = [t_0, T] \times K$ and a compact jet set containing all gradients and Hessians used below. On Q , the payoff under discussion has a decoupling field u that is either $C^{1,2}$ or the locally uniform limit of smooth uniformly parabolic regularizations. The coefficients D, H and their first p -derivatives are bounded and continuous on this jet set. In the nondegenerate diffusion and BSDE statements, also assume $a^u = 2D(x, \nabla u)$ is uniformly positive definite and has a chosen square root σ^u with the regularity needed for the corresponding stochastic equations.

3. LINEARITY OBSTRUCTION

Theorem 3.1 (Linearity obstruction). *Let $\mathcal{G}$ be an operator on a vector space of smooth tests. If there is a single probability law under which*

$$(3) \quad f(t, X_t) - f(0, X_0) - \int_0^t \mathcal{G}f(s, X_s) ds$$

is a martingale for every test f , then $\mathcal{G}$ is additive on that class. Consequently a genuinely nonlinear operator

$$(4) \quad \mathcal{G}f = \partial_t f + \operatorname{tr}(D(x, \nabla f) \nabla^2 f) + H(x, \nabla f)$$

cannot be the generator of one classical martingale problem for all tests.

Proof. Apply the martingale property to f , g , and $f + g$. Subtracting the three martingales gives a finite-variation martingale with drift

$$\mathcal{G}(f + g) - \mathcal{G}f - \mathcal{G}g.$$

It is constant, so the drift vanishes along the support of the law. Varying the initial point gives additivity on the test class. Nonlinear dependence on ∇f or $\nabla^2 f$ violates this except in accidentally linear cases. $\square$

4. PAYOFF-DEPENDENT LINEARIZATION

Definition 4.1 (Calibrated linear generator). Work on a representation window satisfying Assumption 2.3. For a fixed terminal payoff ϕ , let u be its decoupling field. Set

$$(5) \quad a^u(t, x) = 2D(x, \nabla u(t, x)),$$

$$(6) \quad b_k^u(t, x) = \partial_{p_k} H(x, \nabla u(t, x)) + \sum_{i,j} \partial_{p_k} D_{ij}(x, \nabla u(t, x)) \partial_{ij} u(t, x).$$

The payoff-calibrated generator is

$$(7) \quad \mathcal{L}_t^u f(x) = \frac{1}{2} \operatorname{tr}(a^u(t, x) \nabla^2 f(x)) + b^u(t, x) \cdot \nabla f(x).$$

Its carre-du-champ is

$$(8) \quad \Gamma_t^u(f, g) = \frac{1}{2}(\mathcal{L}_t^u(fg) - f\mathcal{L}_t^u g - g\mathcal{L}_t^u f) = \frac{1}{2}\nabla f^\top a^u \nabla g.$$

Theorem 4.2 (Linearized representation for a fixed payoff). *Let u be the smooth solution of (1) with terminal payoff ϕ . If u^δ solves the same HJB equation with terminal payoff $\phi + \delta\eta$ and $u^\delta = u + \delta v + o(\delta)$ locally uniformly, then v solves*

$$(9) \quad -\partial_t v - \mathcal{L}_t^u v = 0, \quad v(T, \cdot) = \eta.$$

Thus the first variation of the nonlinear semigroup at ϕ is represented by the linear parabolic equation generated by $\mathcal{L}^u$.

Proof. Insert $u^\delta = u + \delta v + o(\delta)$ into the HJB operator and differentiate at $\delta = 0$. The derivative of the Hessian term gives $\text{tr}(D(x, \nabla u) \nabla^2 v)$ plus the drift correction

$$\sum_{i,j,k} \partial_{p_k} D_{ij}(x, \nabla u) \partial_{ij} u \partial_k v.$$

The derivative of H gives $\partial_p H(x, \nabla u) \cdot \nabla v$. These terms are exactly $\mathcal{L}_t^u v$ with the sign convention above. $\square$

5. CALIBRATED DIFFUSION REPRESENTATION

Theorem 5.1 (Smooth nondegenerate calibrated diffusion). *Assume a^u is uniformly positive definite and the coefficients in Definition 4.1 are regular enough for the martingale problem of $\mathcal{L}^u$ to be well posed. Then the derivative semigroup in Theorem 4.2 has the classical diffusion representation*

$$(10) \quad dX_s = b^u(s, X_s) ds + \sigma^u(s, X_s) dW_s, \quad \sigma^u(\sigma^u)^\top = a^u.$$

The representing law depends on the payoff through u .

Proof. For fixed u , the operator $\mathcal{L}^u$ is linear. The standard parabolic martingale-problem representation [6, 2] applies to that linear operator. The dependence on ϕ enters through the coefficients a^u and b^u . $\square$

Theorem 5.2 (Degenerate representation by viscosity regularization). *Suppose the viscosity solution u is the locally uniform limit of smooth solutions u^ε to uniformly parabolic regularizations with D replaced by $D + \varepsilon I$. The corresponding calibrated linear diffusion semigroups converge locally uniformly to the payoff-calibrated representation of the original degenerate HJB semigroup.*

Proof. For each $\varepsilon > 0$, the matrix $2(D + \varepsilon I)$ is uniformly positive and gives a classical diffusion representation. Stability of viscosity solutions [1, 2] and stability of the linearized coefficients along the smooth regularization identify the locally uniform limit with the derivative of the original nonlinear semigroup. The construction starts after the effective HJB has been derived. $\square$

6. BSDE REPRESENTATION

Lemma 6.1 (Gradient variable versus BSDE integrand). *In the representation window, the HJB gradient variable is*

$$p(t, x) = \nabla u(t, x).$$

The BSDE martingale integrand is not p . It is

$$Z_t = \sigma(X_t, p(t, X_t))^\top p(t, X_t).$$

When $\sigma^\top$ is invertible on the chosen nondegenerate jet region, one may recover $p = (\sigma^\top)^{-1}Z$. All BSDE drivers below use this conversion only after the payoff and decoupling field are fixed.

Theorem 6.2 (Decoupled FBSDE for a smooth payoff). *Let u be a smooth HJB solution and assume*

$$\sigma(x, p)\sigma(x, p)^\top = 2D(x, p)$$

is nondegenerate on the relevant jet range. Let X solve

$$(11) \quad dX_t = b^u(t, X_t) dt + \sigma(X_t, \nabla u(t, X_t)) dW_t.$$

Set

$$Y_t = u(t, X_t), \quad Z_t = \sigma(X_t, \nabla u(t, X_t))^\top \nabla u(t, X_t).$$

Then (X, Y, Z) satisfies the decoupled BSDE

$$(12) \quad dY_t = -g(t, X_t, Z_t) dt + Z_t dW_t, \quad Y_T = \phi(X_T),$$

where, writing $p = (\sigma^\top)^{-1}Z$,

$$(13) \quad g(t, x, Z) = H(x, p) - b^u(t, x) \cdot p.$$

Proof. Apply Ito's formula to $Y_t = u(t, X_t)$, using the convention in Lemma 6.1. The martingale coefficient is $Z_t = \sigma^\top \nabla u$. The drift is

$$\partial_t u + b^u \cdot \nabla u + \frac{1}{2} \text{tr}(\sigma \sigma^\top \nabla^2 u) = \partial_t u + b^u \cdot p + \text{tr}(D \nabla^2 u).$$

Using the HJB equation, this equals $b^u \cdot p - H(x, p) = -g(t, x, Z)$. $\square$

7. SIGN CONVENTION AND PARABOLIC ORIENTATION

Proposition 7.1 (Parabolic orientation ledger). *The terminal-value convention in Equation (1), the calibrated diffusion orientation, and the BSDE sign convention are consistent as follows. For*

$$Y_s = u(s, X_s), \quad Z_s = \sigma(X_s, \nabla u(s, X_s))^\top \nabla u(s, X_s),$$

the differential form

$$dY_s = -g(s, X_s, Z_s) ds + Z_s dW_s$$

is equivalent to the backward integral equation

$$Y_t = \phi(X_T) + \int_t^T g(s, X_s, Z_s) ds - \int_t^T Z_s dW_s.$$

If one writes forward parabolic time as $\tau = T - t$ and $U(\tau, x) = u(T - \tau, x)$, then the same terminal-value HJB becomes

$$\partial_\tau U - \text{tr}(D(x, \nabla U) \nabla^2 U) - H(x, \nabla U) = 0, \quad U(0, \cdot) = \phi.$$

Thus no sign is changed when passing from the terminal HJB to the backward BSDE; only the time orientation changes.

Proof. For the calibrated diffusion, Ito's formula gives

$$dY_s = (\partial_s u + b^u \cdot \nabla u + \text{tr}(D(x, \nabla u) \nabla^2 u))(s, X_s) ds + Z_s dW_s.$$

The terminal-value HJB gives

$$\partial_s u + \text{tr}(D(x, \nabla u) \nabla^2 u) = -H(x, \nabla u),$$

so the drift is $b^u \cdot p - H(x, p) = -g(s, x, Z)$. Integrating the differential identity from t to T gives the displayed BSDE. The forward-time equation follows by the substitution $U(\tau, x) = u(T - \tau, x)$, for which $\partial_\tau U = -\partial_t u$. $\square$

8. NONLINEAR CAMERON-MARTIN-GIRSANOV FORMULA

Definition 8.1 (Hamiltonian shift). For a deterministic shift $h(t, x)$ define

$$(14) \quad H^h(x, p) = H(x, p + h) - H(x, h),$$

$$(15) \quad D^h(x, p) = D(x, p + h).$$

The shifted semigroup is generated by the HJB equation with D, H replaced by D^h, H^h .

Theorem 8.2 (Semigroup-level Cameron-Martin formula). *Let u_h solve the shifted equation. Then u_h also solves the unshifted equation with deterministic generator defect*

$$(16) \quad \Lambda_h(t, x, p, X) = \text{tr}((D(x, p + h) - D(x, p))X) + H(x, p + h) - H(x, h) - H(x, p).$$

Equivalently,

$$(17) \quad -\partial_t u_h - \text{tr}(D(x, \nabla u_h) \nabla^2 u_h) - H(x, \nabla u_h) - \Lambda_h(t, x, \nabla u_h, \nabla^2 u_h) = 0.$$

When D is independent of p and H is linear-quadratic, this identity reduces after linearization to the classical Cameron-Martin-Girsanov drift tilt [5].

Proof. Insert the definitions of D^h and H^h into the shifted HJB equation. Add and subtract the unshifted terms

$$\text{tr}(D(x, \nabla u_h) \nabla^2 u_h) + H(x, \nabla u_h).$$

The remainder is exactly (16). The linear-quadratic special case gives $H(x, p + h) - H(x, h) - H(x, p) = p^\top a(x)h$, the usual drift-tilt term for the payoff-calibrated linear diffusion. $\square$

9. A SHORT ROBUST-PRICING ILLUSTRATION

Definition 9.1 (Calibrated stress family). Fix a terminal payoff ϕ in a representation window and let $\mathfrak{H}$ be a compact family of deterministic shifts $h(t, x)$ for which the shifted equations in Definition 8.1 remain in the same regularity window. For each $h \in \mathfrak{H}$, let u^h be the shifted solution with terminal value ϕ . The calibrated stress envelope is

$$\Pi_{\mathfrak{H}}\phi(t, x) = \sup_{h \in \mathfrak{H}} u^h(t, x).$$

This envelope is a post-derivation stress test for the already constructed theta-expectation semigroup, not a new primitive market model.

Proposition 9.2 (Robust-pricing stress test). *In the smooth nondegenerate representation window, each $h \in \mathfrak{H}$ has a payoff-calibrated diffusion and BSDE representation obtained from the shifted coefficients D^h, H^h . The envelope $\Pi_{\mathfrak{H}}\phi$ therefore records a family of calibrated drift and Hamiltonian stresses for the fixed payoff. When D is independent of p and H is linear-quadratic, this reduces to the familiar drift-uncertainty stress test after the Cameron-Martin-Girsanov tilt. In the nonconvex theta-expectation case the envelope should not be read as a sublinear expectation unless the additional sublinearity inequalities are separately verified.*

Proof. For each fixed h , Theorem 8.2 rewrites the shifted equation as the unshifted equation plus the deterministic defect Λ_h . Once the payoff and shifted decoupling field u^h are fixed, the construction of Definition 4.1 and Theorems 5.1 and 6.2 applies with D, H, u replaced by D^h, H^h, u^h . Compactness of $\mathfrak{H}$ and the representation-window bounds keep the family in one local regularity class. In the linear-quadratic case the defect is the classical drift tilt $p^\top ah$. For a nonconvex Hamiltonian, taking a supremum over calibrated stresses gives a useful pricing diagnostic, but it does not restore the global subadditivity that Paper 2 proves can fail. $\square$

Proposition 9.3 (Post-derivation calibration). *The calibrated laws and BSDEs in Theorems 5.1, 5.2 and 6.2 are downstream representations of a fixed payoff. They do not form a single payoff-independent martingale problem and cannot be used to prove Assumption 2.1.*

Proof. The law in Theorem 5.1 depends on u , and u is determined by the terminal payoff through the nonlinear HJB semigroup. Changing the payoff changes u , hence changes a^u , b^u , the calibrated diffusion, and the BSDE driver. A payoff-independent law would contradict Theorem 3.1 except in accidentally linear cases. Therefore the logical order is acyclic: Paper 2 first supplies the semigroup, then the present paper represents its fixed-payoff derivatives. $\square$

10. MAIN REPRESENTATION PACKAGE

Theorem 10.1 (Representation calculus package). *Under Assumption 2.1, every sufficiently regular terminal payoff determines:*

- (i) *a payoff-calibrated linearized generator $\mathcal{L}^u$;*
- (ii) *a carre-du-champ Γ^u ;*
- (iii) *a smooth or regularized calibrated diffusion representation;*
- (iv) *a decoupled BSDE representation on nondegenerate jet regions;*
- (v) *a semigroup-level Cameron–Martin–Girsanov formula.*
- (vi) *a terminal-value sign-convention ledger;*
- (vii) *for any compact deterministic stress family, a calibrated robust-pricing stress envelope.*

All objects above are post-derivation representations of the theta-expectation semigroup. None is used to prove the deterministic billiard response theorem or the HJB homogenization theorem.

Proof. Items (i) and (ii) are Definition 4.1. Item (iii) is Theorems 5.1 and 5.2. Item (iv) is Theorem 6.2. Item (v) is Theorem 8.2. Item (vi) is Proposition 7.1. Item (vii) is Definition 9.1 and Proposition 9.2. The downstream character follows from the single input assumption Assumption 2.1 and the calibration guardrail in Proposition 9.3. $\square$

11. SUBMISSION SCOPE

The paper’s submission version should keep smoothness, nondegeneracy, and localization assumptions close to each representation theorem. It should not import billiard technical lemmas except through Paper 2’s HJB theorem. The robust-pricing example above should remain a short illustration of calibrated stress testing, not an additional primitive derivation.

REFERENCES

- [1] M. G. Crandall, H. Ishii, and P.-L. Lions. User’s guide to viscosity solutions of second order partial differential equations. *Bulletin of the American Mathematical Society*, 27(1):1–67, 1992.
- [2] W. H. Fleming and H. M. Soner. *Controlled Markov Processes and Viscosity Solutions*. Springer, 2nd edition, 2006.
- [3] É. Pardoux and S. Peng. Backward stochastic differential equations and quasilinear parabolic partial differential equations. In B. L. Rozovskii and R. B. Sowers, editors, *Stochastic Partial Differential Equations and Their Applications*, volume 176 of *Lecture Notes in Control and Information Sciences*, pages 200–217. Springer, 1992.
- [4] S. Peng. *Nonlinear Expectations and Martingale Theory in Mathematical Finance*. Springer, 2019.
- [5] Philip E. Protter. *Stochastic Integration and Differential Equations*, volume 21 of *Stochastic Modelling and Applied Probability*. Springer, Berlin, 2012.
- [6] Daniel W. Stroock and S. R. S. Varadhan. *Multidimensional Diffusion Processes*. Classics in Mathematics. Springer, Berlin, 2007.